Problem 1. *Recall the nonparametric selection model discussed in class:*

$$\begin{aligned} y^* &= m(x) + \epsilon \\ d &= \mathbf{1}\{p(x,z) \geq u\} \\ y &= dy^* \end{aligned}$$

with $(\epsilon, u) \perp (x, z)$*, and* $u|x, z \sim$ *Unif*$(0,1)$*. The data* (x, y, z, d) *are observed. The functions* $m : \mathbb{R} \to \mathbb{R}$*,* $p : \mathbb{R}^2 \to \mathbb{R}$ *and the joint distribution of* (ϵ, u) *are all unknown, though it is trivial to identify* $p(\cdot, \cdot)$*. Fix two points* $\bar{x}, x \in \mathbb{R}$*. The objective is to identify the slope*

$$\frac{m(\bar{x}) - m(x)}{\bar{x} - x}. \tag{1}$$

(i) An assumption is now imposed (maintained for the remainder of this question): suppose two values $\bar{z}, z \in \mathbb{R}$ *can be found such that*

$$p(\bar{x}, \bar{z}) = p(x, z).$$

Discuss what kind of conditions make this assumption plausible.

(ii) Use the Wald-like approach to obtain an expression for the slope (1) in terms of observables.

(iii) Verify that by directly differentiating $E[y|x^*, z^*, d = 1]$ *with respect to* x^* *and* z^* *and assuming an appropriate rank condition, the following formula for the structural derivative of* m *at* $x^* \in \mathbb{R}$ *can be derived:*

$$\frac{\partial}{\partial x} m(x^*) = \frac{\partial}{\partial x^*} E[y|x^*, z^*, d = 1] - \frac{\frac{\partial}{\partial z^*} E[y|x^*, z^*, d = 1] \frac{\partial}{\partial x^*} p(x^*, z^*)}{\frac{\partial}{\partial z^*} p(x^*, z^*)},$$

where $z^* \in \mathbb{R}$*.*

(iv) Take the limit $\bar{x}, x \to x^*$ *(and* $\bar{z}, z \to z^*$*) in the answer from (ii) to show that the derivative formula in (iii) is replicated through the Wald-like approach. (Hint: Since* $p(\bar{x}, \bar{z}) = p(x, z)$*, the expression*

$$\frac{p(\bar{x}, \bar{z}) - p(x, \bar{z})}{\bar{x} - x} + \frac{p(x, \bar{z}) - p(x, z)}{\bar{x} - x}$$

(after multiplying the second term by $\frac{\bar{z} - z}{\bar{z} - z}$*) being zero implies*

$$\frac{\partial}{\partial x^*} p(x^*, z^*) + \frac{\partial}{\partial z^*} p(x^*, z^*) \lim \frac{\bar{z} - z}{\bar{x} - x} = 0.)$$

Solution.

(i) This assumption is plausible if the instrument z affects the selection probability $p(x, z)$ in a way that can offset changes in x. This requires instrument relevance ($\partial p / \partial z \neq 0$) and sufficient variation in z such that for any $\bar{x}$ and x, there exist $\bar{z}$ and z satisfying $p(\bar{x}, \bar{z}) = p(x, z)$. Geometrically, this amounts to the level sets of $p(\cdot, \cdot)$ being well-behaved: for any two x-values, the level curve must pass through points with different z-values.

(ii) Under the assumption $p(\bar{x}, \bar{z}) = p(x, z) = p_0$,

$$E[y \mid \bar{x}, \bar{z}] = p_0[m(\bar{x}) + \lambda(p_0)], \quad E[y \mid x, z] = p_0[m(x) + \lambda(p_0)],$$

where $\lambda(p_0) = E[\epsilon \mid p(x, z) = p_0, d = 1]$. Subtracting yields

$$E[y \mid \bar{x}, \bar{z}] - E[y \mid x, z] = p_0[m(\bar{x}) - m(x)].$$

Thus, the slope is identified as

$$\frac{m(\bar{x}) - m(x)}{\bar{x} - x} = \frac{1}{p_0} \frac{E[y \mid \bar{x}, \bar{z}] - E[y \mid x, z]}{\bar{x} - x}.$$

(iii) From $E[y \mid x, z, d = 1] = m(x) + \lambda(p(x, z))$, differentiation with respect to x^* and z^* gives

$$\frac{\partial}{\partial x^*} E[y \mid x^*, z^*, d = 1] = m'(x^*) + \lambda'(p) \frac{\partial p}{\partial x^*},$$

$$\frac{\partial}{\partial z^*} E[y \mid x^*, z^*, d = 1] = \lambda'(p) \frac{\partial p}{\partial z^*}.$$

Eliminating $\lambda'(p)$ from the second equation and substituting into the first:

$$m'(x^*) = \frac{\partial}{\partial x^*} E[y \mid x^*, z^*, d = 1] - \frac{\frac{\partial}{\partial z^*} E[y \mid x^*, z^*, d = 1] \frac{\partial}{\partial x^*} p(x^*, z^*)}{\frac{\partial}{\partial z^*} p(x^*, z^*)}.$$

(iv) From part (ii), the Wald-like estimator is

$$\frac{m(\bar{x}) - m(x)}{\bar{x} - x} = \frac{E[y \mid \bar{x}, \bar{z}] - E[y \mid x, z]}{p_0(\bar{x} - x)}.$$

As $(\bar{x}, x) \to (x^*, x^*)$ and $(\bar{z}, z) \to (z^*, z^*)$, the numerator becomes

$$\frac{\partial}{\partial x^*} E[y \mid x^*, z^*](\bar{x} - x) + \frac{\partial}{\partial z^*} E[y \mid x^*, z^*](\bar{z} - z) + o(|\bar{x} - x| + |\bar{z} - z|).$$

Using the hint, since $p(\bar{x}, \bar{z}) = p(x, z)$, expanding both sides:

$$\frac{\partial p}{\partial x^*}(x^*, z^*) + \frac{\partial p}{\partial z^*}(x^*, z^*) \lim_{(\bar{x}, \bar{z}, x, z) \to (x^*, z^*, x^*, z^*)} \frac{\bar{z} - z}{\bar{x} - x} = 0,$$

which implies

$$\lim \frac{\bar{z} - z}{\bar{x} - x} = -\frac{\partial p / \partial x^*}{\partial p / \partial z^*}.$$

Substituting this into the limit of the Wald estimator:

$$m'(x^*) = \frac{1}{p_0} \left[\frac{\partial}{\partial x^*} E[y \mid x^*, z^*] - \frac{\partial}{\partial z^*} E[y \mid x^*, z^*] \frac{\partial p / \partial x^*}{\partial p / \partial z^*} \right].$$

Since $p_0 = p(x^*, z^*)$, and using $E[y \mid x^*, z^*] = p(x^*, z^*) E[y \mid x^*, z^*, d = 1]$, the formula from (iii) is recovered.

□

Problem 2. *Consider a nonparametric model with an endogenous regressor:*

$$y = m(x) + \epsilon$$
$$x = \pi(z, \eta)$$

where $m : \mathbb{R} \to \mathbb{R}$, $\pi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, *and*

$$E[\epsilon | z, \eta] = C(\eta)$$

(i.e., the expectation of ϵ *conditional on* z *and* η *does not depend on* z*). Suppose* $z \perp \eta$*. Additionally,* m*,* π *and* C *are all differentiable in their arguments, and* π *is strictly increasing in* η *for every value of* z*. Normalize* η *so that*

$$\eta | z \sim \textit{Unif}(0, 1).$$

(i) Consider two values $\bar{z}_0, \bar{z}_1$ *for* z *and a value* α *between 0 and 1 for* η*. Discuss the identifiability of the slope of* m *using the Wald-type argument. What would be an appropriate rank condition?*

(ii) Obtain the derivative of m *at* $\pi(\bar{z}, \alpha)$ *for some* $\bar{z}$ *and* $0 < \alpha < 1$ *under the rank condition* $\frac{\partial \pi(\bar{z}, \alpha)}{\partial z} \neq 0$*.*

Solution.

(i) For fixed $\eta = \alpha$, let $x_0 = \pi(\bar{z}_0, \alpha)$ and $x_1 = \pi(\bar{z}_1, \alpha)$. Then

$$E[y \mid \bar{z}_0, \eta = \alpha] = m(x_0) + C(\alpha), \quad E[y \mid \bar{z}_1, \eta = \alpha] = m(x_1) + C(\alpha).$$

Subtracting yields

$$E[y \mid \bar{z}_1, \eta = \alpha] - E[y \mid \bar{z}_0, \eta = \alpha] = m(x_1) - m(x_0).$$

Thus, the slope is

$$\frac{m(x_1) - m(x_0)}{x_1 - x_0} = \frac{E[y \mid \bar{z}_1, \eta = \alpha] - E[y \mid \bar{z}_0, \eta = \alpha]}{\pi(\bar{z}_1, \alpha) - \pi(\bar{z}_0, \alpha)}.$$

The rank condition is $\pi(\bar{z}_1, \alpha) \neq \pi(\bar{z}_0, \alpha)$, which ensures the denominator is nonzero and the first stage has sufficient variation.

(ii) The derivative of m at $\pi(\bar{z}, \alpha)$ is the limit of the Wald ratio as $\bar{z}_1 \to \bar{z}$:

$$m'(\pi(\bar{z}, \alpha)) = \lim_{\bar{z}_1 \to \bar{z}} \frac{E[y \mid \bar{z}_1, \eta = \alpha] - E[y \mid \bar{z}, \eta = \alpha]}{\pi(\bar{z}_1, \alpha) - \pi(\bar{z}, \alpha)} = \frac{\frac{\partial}{\partial z} E[y \mid z, \eta = \alpha]\big|_{z=\bar{z}}}{\frac{\partial}{\partial z} \pi(z, \alpha)\big|_{z=\bar{z}}}.$$

Under the rank condition $\frac{\partial \pi(\bar{z}, \alpha)}{\partial z} \neq 0$, this derivative is well-defined and identifies $m'(\pi(\bar{z}, \alpha))$.

□

Problem 3. *Consider random variables x, y and z taking values in $\mathbb{R}^k$, $\mathbb{R}$ and $\mathbb{R}$, respectively.*

(i) Let $z \sim F_z$. Define

$$y = \max\{0, z\}$$

and let F_y denote the CDF of y. Only y is observed, not z. Let $\mu_z = E[z] = \int z \, dF_z$. Can μ_z be identified from the distribution F_y?

(ii) Consider the following model:

$$y = \max\{0, z\}, \quad z = m(x) + \epsilon.$$

Here $m(\cdot)$ is a (nonparametric) function and $F_{\epsilon|x}$ denotes the conditional CDF of ϵ given x. Suppose $E[\epsilon|x] = 0$ is assumed. Only x and y are observed, not z. Is the function m (or some feature of it) identified from the knowledge of the joint distribution of x and y? What if $Q_{\epsilon|x}(\tau|x) = 0$ is assumed instead of the conditional mean zero restriction ($\tau \in (0,1)$ is a known constant)? Is the function m (or some feature of it) identified? What if $Q_{\epsilon|x}(\tau|x)$ is assumed to be an unknown constant (that does not depend on x)? Explain.

Solution.

(i) No, μ_z is not identified from F_y. The distribution of z is only observed for $z \geq 0$, while the distribution for $z < 0$ is censored at zero. Since $E[z] = E[z \mid z \geq 0]P(z \geq 0) + E[z \mid z < 0]P(z < 0)$, and $E[z \mid z < 0]$ cannot be recovered from F_y, the mean μ_z is not identified.

(ii) Three cases are considered:

- **Under $E[\epsilon|x] = 0$:** Since $E[z|x] = m(x)$ and $y = \max\{0, z\}$,

$$E[y|x] = E[\max\{0, m(x) + \epsilon\} \mid x] \neq m(x)$$

in general due to censoring. The function $m(x)$ is not identified from $E[y|x]$ because the censoring introduces a nonlinear transformation that depends on the unknown distribution of $\epsilon|x$.

- **Under $Q_{\epsilon|x}(\tau|x) = 0$:** The τ-th conditional quantile of z given x is $Q_{z|x}(\tau|x) = m(x)$. For the censored variable $y = \max\{0, z\}$,

$$Q_{y|x}(\tau|x) = \max\{0, Q_{z|x}(\tau|x)\} = \max\{0, m(x)\}.$$

Thus, $m(x)$ is identified only for x such that $m(x) \geq 0$. When $m(x) < 0$, the quantile is censored at zero and $m(x)$ cannot be recovered.

- **Under $Q_{\epsilon|x}(\tau|x) = c$ (unknown constant):** Then $Q_{z|x}(\tau|x) = m(x) + c$, and

$$Q_{y|x}(\tau|x) = \max\{0, m(x) + c\}.$$

Thus, $m(x) + c$ is identified where $m(x) + c \geq 0$. The function $m(x)$ is identified only up to the additive constant c. If there exists some x_0 where $m(x_0) + c$ is known or can be normalized (e.g., if $m(x_0) + c > 0$ is known from the data), then c can be identified and hence $m(x)$ for all x where $m(x) + c \geq 0$.

□

Problem 4. *Consider the following nonparametric regression model with endogeneity and multiplicative heteroskedasticity of unknown form:*

$$\begin{aligned} y &= g_0(x) + h_0(z_1) \times \epsilon \\ x &= \Pi_0(z) + \eta, \quad z = (z_1', z_2')' \\ E[\epsilon|\eta, z] &= c_0(\eta) \\ E[\eta|z] &= 0. \end{aligned}$$

Let d_x, d_z, d_{z_1} and d_{z_2} denote the dimensionality of the vectors x, z, z_1 and z_2, respectively. The functions $g_0 : \mathbb{R}^{d_x} \to \mathbb{R}$, $h_0 : \mathbb{R}^{d_{z_1}} \to \mathbb{R}$, $\Pi_0 : \mathbb{R}^{d_z} \to \mathbb{R}^{d_x}$ and $c_0 : \mathbb{R}^{d_x} \to \mathbb{R}$ are unknown. How would the structural derivative $\nabla_x g_0(x)$ on $\mathbb{R}^{d_x}$ be identified? Assume conditions analogous to those used in Newey, Powell, Vella (1999), that is,

$$rank \frac{\partial \Pi_0(z)}{\partial z_2'} = d_x \text{ for every } z.$$

Also, assume that g_0 and Π_0 (and h_0 and c_0 as well) are differentiable everywhere over $\mathbb{R}^{d_x}$ and $\mathbb{R}^{d_z}$, and the support of $(\eta', z')'$ is $\mathbb{R}^{d_x} \times \mathbb{R}^{d_z}$.

Solution. Let $\eta = x - \Pi_0(z)$. Then

$$E[y \mid x, z] = g_0(x) + h_0(z_1)c_0(\eta).$$

Note that $\nabla_x g_0(x) \in \mathbb{R}^{d_x}$ is a d_x-dimensional gradient vector, and all derivatives below should be interpreted as vectors or matrices of appropriate dimensions.

Differentiate with respect to z_2 (the $d_{z_2} \times 1$ vector of excluded instruments):

$$\frac{\partial E[y \mid x, z]}{\partial z_2'} = h_0(z_1)c_0'(\eta)\frac{\partial \eta}{\partial z_2'} = -h_0(z_1)c_0'(\eta)\frac{\partial \Pi_0(z)}{\partial z_2'},$$

where $\frac{\partial E[y|x,z]}{\partial z_2'}$ is a $1 \times d_{z_2}$ vector and $\frac{\partial \Pi_0(z)}{\partial z_2'}$ is a $d_x \times d_{z_2}$ matrix. Also, $c_0'(\eta)$ should be interpreted as $\nabla_\eta c_0(\eta)$, a $1 \times d_x$ row vector.

From the reduced form $x = \Pi_0(z)$,

$$\frac{\partial x}{\partial z_2'} = \frac{\partial \Pi_0(z)}{\partial z_2'}.$$

Under the rank condition, $\frac{\partial \Pi_0(z)}{\partial z_2'}$ has full row rank d_x. Therefore,

$$h_0(z_1)c_0'(\eta) = -\frac{\partial E[y \mid x, z]}{\partial z_2'}\left(\frac{\partial \Pi_0(z)}{\partial z_2'}\right)^{-R},$$

where $(\cdot)^{-R}$ denotes a right inverse (since the matrix is $d_x \times d_{z_2}$ with $d_{z_2} \geq d_x$).

Now differentiate $E[y \mid x, z]$ with respect to x (a $d_x \times 1$ vector):

$$\frac{\partial E[y \mid x, z]}{\partial x'} = \nabla_x g_0(x)' + h_0(z_1)c_0'(\eta)\frac{\partial \eta}{\partial x'} = \nabla_x g_0(x)' + h_0(z_1)c_0'(\eta),$$

where $\nabla_x g_0(x)$ is a $d_x \times 1$ gradient vector and $\frac{\partial E[y|x,z]}{\partial x'}$ is a $1 \times d_x$ row vector.
Solving for $\nabla_x g_0(x)$:

$$\nabla_x g_0(x) = \left[\frac{\partial E[y \mid x, z]}{\partial x'}\right]' - \left[\frac{\partial E[y \mid x, z]}{\partial z_2'}\left(\frac{\partial x}{\partial z_2'}\right)^{-R}\right]'.$$

Alternatively, using matrix notation more carefully:

$$\nabla_x g_0(x) = \nabla_x E[y \mid x, z] + \left[\frac{\partial \Pi_0(z)}{\partial z_2'}\right]^{-R\top} \nabla_{z_2} E[y \mid x, z],$$

where $\nabla_x E[y \mid x, z]$ is the $d_x \times 1$ gradient with respect to x and $\nabla_{z_2} E[y \mid x, z]$ is the $d_{z_2} \times 1$ gradient with respect to z_2.

This identifies the structural derivative $\nabla_x g_0(x)$ at each point x in terms of observable conditional expectations and their derivatives. □

Problem 5. *Suppose data $\{y_i, x_i, z_{1i}, z_{2i}\}_{i=1}^N$ are observed, where y_i, x_i, z_{1i}, z_{2i} take values in $\mathbb{R}, \mathbb{R}^{d_x}, \mathbb{R}^{d_{z_1}}$, and $\mathbb{R}^{d_{z_2}}$, respectively (for every i). Let $z_i = [z_{1i}^\top, z_{2i}^\top]^\top$. The objective is to estimate a linear model*

$$y_i = x_i^\top \beta + z_{1i}^\top \gamma + \epsilon_i = w_i^\top \theta + \epsilon_i, \quad w_i = [x_i^\top, z_{1i}^\top]^\top$$

where x_i is endogenous, using z_i as the vector of instruments. Standard vector/matrix notation is employed:

$$y = [y_1, ..., y_N]^\top, \quad X = [x_1, ..., x_N]^\top, \quad Z_1 = [z_{11}, ..., z_{1N}]^\top, \quad Z_2 = [z_{21}, ..., z_{2N}]^\top,$$

$Z = [Z_1, Z_2]$, $W = [X, Z_1]$. The notation for the projection matrix is

$$P_Z = Z(Z^\top Z)^{-1} Z^\top.$$

The standard TSLS estimator for θ is then

$$\hat{\theta} = [W^\top Z(Z^\top Z)^{-1} Z^\top W]^{-1} W^\top Z(Z^\top Z)^{-1} Z^\top y = [W^\top P_Z W]^{-1} W^\top P_Z y.$$

Consider now a control function (CF) approach. Let

$$\hat{\Pi} = [\hat{\Pi}_1, \hat{\Pi}_2] = (Z^\top Z)^{-1} Z^\top X$$

and

$$\hat{\eta} = X - Z\hat{\Pi} = (I - P_Z)X.$$

The i-th row of $\hat{\eta}$ is the estimated control variable for individual i. Regress y on W and $\hat{\eta}$:

$$\begin{pmatrix} \tilde{\theta} \\ \tilde{\lambda} \end{pmatrix} = [F^\top F]^{-1} F^\top y, \quad F = [W, \hat{\eta}].$$

Show that in such a linear case

$$\tilde{\theta} = \hat{\theta}.$$

Solution. By the Frisch-Waugh-Lovell theorem, the coefficient on W in the regression of y on $[W, \hat{\eta}]$ is

$$\tilde{\theta} = [W^\top (I - P_{\hat{\eta}})W]^{-1} W^\top (I - P_{\hat{\eta}})y,$$

where $P_{\hat{\eta}} = \hat{\eta}(\hat{\eta}^\top \hat{\eta})^{-1}\hat{\eta}^\top$ is the projection matrix onto the column space of $\hat{\eta}$.

Note that $\hat{\eta} = (I - P_Z)X$ is orthogonal to the column space of Z, since

$$Z^\top \hat{\eta} = Z^\top (I - P_Z)X = (Z^\top - Z^\top P_Z)X = (Z^\top - Z^\top)X = 0.$$

Since $W = [X, Z_1]$ and Z_1 is a submatrix of $Z = [Z_1, Z_2]$, both Z_1 and $X - P_Z X$ are in the column space of $[Z, (I - P_Z)X]$.

For any vector in the column space of Z, $(I - P_{\hat{\eta}})$ has no effect beyond what P_Z already captures. More precisely, since $(I - P_Z)W = (I - P_Z)[X, Z_1] = [(I - P_Z)X, 0]$ and $\hat{\eta} = (I - P_Z)X$, the residual maker $I - P_{\hat{\eta}}$ removes the component in the direction of $(I - P_Z)X$.

The key insight is that $W^\top (I - P_{\hat{\eta}}) = W^\top P_Z (I - P_{\hat{\eta}})$, because:

$$W^\top (I - P_Z)(I - P_{\hat{\eta}}) = [X, Z_1]^\top (I - P_Z)(I - P_{\hat{\eta}}) = [(I - P_Z)X, 0]^\top (I - P_{\hat{\eta}}) = 0,$$

since $(I - P_Z)X = \hat{\eta}$ is in the column space of $\hat{\eta}$, so $(I - P_{\hat{\eta}})\hat{\eta} = 0$.

Therefore,

$$W^\top (I - P_{\hat{\eta}})W = W^\top [P_Z + (I - P_Z)](I - P_{\hat{\eta}})W = W^\top P_Z (I - P_{\hat{\eta}})W = W^\top P_Z W,$$

where the last equality follows because $P_Z W$ is in the column space of Z, which is orthogonal to $\hat{\eta}$, so $(I - P_{\hat{\eta}})P_Z W = P_Z W$.

Similarly,

$$W^\top (I - P_{\hat{\eta}})y = W^\top P_Z (I - P_{\hat{\eta}})y = W^\top P_Z y.$$

Thus,

$$\tilde{\theta} = [W^\top P_Z W]^{-1} W^\top P_Z y = \hat{\theta}.$$

□